

DEWAR LAKES, NU, Canada

WMO: 710920

Lat: 68.651N

Lon: 71.234W

Elev: 527

StdP: 95.16

Time Zone: -5.00 (NAE)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-39.5	-37.7	-40.7	0.1	-36.7	-40.0	0.1	-36.4	20.7	-17.6	18.3	-21.6	3.8	290	1.150

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.3	14.9	9.1	12.8	8.1	11.0	7.2	9.3	13.5	8.3	11.6	7.3	10.0	4.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.1	6.7	9.3	6.0	6.2	9.2	5.0	5.8	8.3	29.0	13.4	26.5	11.7	24.3	10.1	12.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.8	14.5	12.8	DB	-40.8	16.6	2.2	2.8	-42.4	18.7	-43.7	20.3	-44.9	21.9	-46.5	24.0	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-12.0	-26.9	-27.7	-24.9	-18.1	-8.5	0.4	6.9	4.8	-2.0	-9.5	-16.8	-22.6
	DBStd	13.05	6.34	6.36	6.76	5.73	4.72	3.54	3.61	3.37	3.27	4.59	5.84	6.57
	HDD10.0	8041	1143	1054	1083	842	573	287	111	166	362	604	804	1012
	HDD18.3	11065	1402	1288	1341	1092	831	537	356	420	612	862	1054	1271
	CDD10.0	18	0	0	0	0	0	0	14	4	0	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	5.7	5.3	5.3	5.4	5.7	6.0	5.9	5.6	5.6	5.6	5.7	6.1	5.6
Precipitation	PrecAvg	248	5	3	5	9	20	19	46	64	34	25	13	6
	PrecMax	405	30	16	47	33	50	68	103	153	73	68	42	27
	PrecMin	107	0	0	0	0	2	0	3	6	8	9	0	0
	PrecStd	86	7	4	9	8	13	17	28	42	17	14	9	6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.2	-7.8	-5.6	-2.0	2.7	11.8	18.6	15.7	6.6	0.3	-2.8	-4.3
		MCWB	-5.2	-7.8	-5.6	-4.7	1.1	7.1	10.8	9.7	4.5	0.3	-2.8	-7.5
	2%	DB	-11.0	-12.5	-9.3	-4.9	0.8	9.3	16.3	13.3	4.6	-1.2	-4.8	-7.5
		MCWB	-11.2	-12.8	-9.9	-6.5	-0.4	5.5	9.7	8.3	3.1	-1.3	-5.1	-8.5
	5%	DB	-15.4	-15.6	-12.4	-7.7	-0.8	7.2	14.3	11.5	3.3	-2.2	-6.4	-11.0
		MCWB	-15.9	-16.0	-13.1	-8.4	-1.4	4.2	8.8	7.7	2.3	-2.3	-6.7	-11.5
	10%	DB	-18.2	-18.8	-15.3	-10.1	-2.2	5.6	12.5	9.9	2.1	-3.5	-8.4	-14.2
		MCWB	-18.6	-19.2	-15.8	-10.7	-2.7	3.3	8.0	6.7	1.5	-3.7	-8.5	-14.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-15.8	-9.5	-9.4	-4.9	3.8	7.7	11.3	9.7	5.1	-0.4	-4.2	-8.5
		MCDB	-15.5	-9.4	-9.1	-2.9	6.8	11.1	17.3	13.5	6.2	-0.4	-4.1	-7.6
	2%	WB	-17.2	-12.4	-12.9	-8.1	0.7	6.3	10.1	8.6	3.2	-1.5	-6.1	-11.5
		MCDB	-16.8	-12.2	-12.4	-7.6	1.8	9.5	15.0	11.8	4.0	-1.5	-5.9	-11.1
	5%	WB	-19.2	-15.7	-15.8	-10.0	-0.4	5.0	9.1	7.8	2.2	-2.4	-7.8	-13.9
		MCDB	-18.8	-15.4	-15.4	-9.6	0.4	8.1	13.2	10.6	2.8	-2.4	-7.5	-13.5
	10%	WB	-20.8	-18.9	-18.4	-12.4	-1.7	3.8	8.0	6.5	1.4	-3.8	-10.2	-15.7
		MCDB	-20.5	-18.6	-18.0	-11.9	-1.1	6.1	11.4	8.8	1.9	-3.7	-10.0	-15.4
Mean Daily Temperature Range	5% DB	MDBR	5.1	5.0	5.1	4.7	3.8	3.8	5.3	4.3	3.0	3.8	4.7	5.0
		MCDBR	7.3	7.5	6.7	5.7	4.3	5.8	7.2	6.2	4.2	3.3	5.6	6.5
		MCWBR	9.3	7.2	6.4	5.3	3.4	3.9	3.8	3.6	3.0	2.8	4.9	4.5
	5% WB	MCDBR	6.7	7.4	6.3	6.2	4.2	5.9	6.8	5.7	3.6	2.7	4.8	4.9
		MCWBR	6.6	7.2	6.2	5.9	3.4	4.1	3.8	3.3	2.9	2.8	4.9	4.8
Clear-Sky Solar Irradiance	taub	0.150	0.196	0.235	0.282	0.310	0.312	0.322	0.309	0.268	0.214	0.148	N/A	
	taud	1.874	2.086	2.232	2.135	2.148	2.358	2.500	2.532	2.464	2.103	1.906	N/A	
	Ebn at Noon	100	626	822	861	872	887	868	842	788	564	86	0	
	Edh at Noon	6	42	71	108	121	102	86	74	59	40	5	0	
All-Sky Solar Radiation	RadAvg	0.03	0.51	2.15	4.56	6.32	5.86	4.88	3.34	1.72	0.55	0.09	0.00	
	RadStd	0.00	0.03	0.08	0.14	0.37	0.88	0.48	0.34	0.16	0.11	0.01	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		+0.78	+1.39	N/A	N/A	N/A	N/A	-282	-281	N/A	N/A

Nomenclature: See separate page